Symmetric PKM: terminal evidence and dependency integration

2026-10-30 · Software validation and G1–G6 prerequisite work

The research question remains open. G1–G3/G6 are partial and G4/G5 are unrun. No scientific panel, priced gain, GAE/scaling result or new cheap relation algorithm is supplied. Comparison admission remains false.

Final outcome on the earlier pinned source

Source faade165a46fd51e5a0903ba8e4df85a134c68db has 23 passing example tests, strict scoped lint/format, CLI build/refusal, backlog validation and library build passes. Its full serial library execution reached the 40-minute cap, 16:47:54–17:27:54 UTC:

Declared	Passed	Failed	Ignored	Active	Not reached
5646	2208	0	47	1	3390

The unchanged JV end-to-end recovery case remained active. Wrapper/supervisor exit 124 is not Cargo's direct exit, which is unknown. There is no final suite summary or full-library pass. The count parser includes should-panic harness annotations. All 17 source objects and six binary hashes matched at collection. Cause of the runtime remains unknown; no deadlock, nontermination or PKM impossibility is inferred.

The 214615-byte raw log is NOT_UPLOADED. SHA256: `34fdd95ea133361298e8911f9a5899c431a5e9af97b73e7828e1b5289cdf87bd`. Exact path/hash/size and a bounded excerpt are retained in validation/native-final-faade/. It is distinct from earlier same-size logs with different hashes. No compression/splitting, ignore, assertion change or gate waiver occurred. Historical receipts and reports remain unchanged.

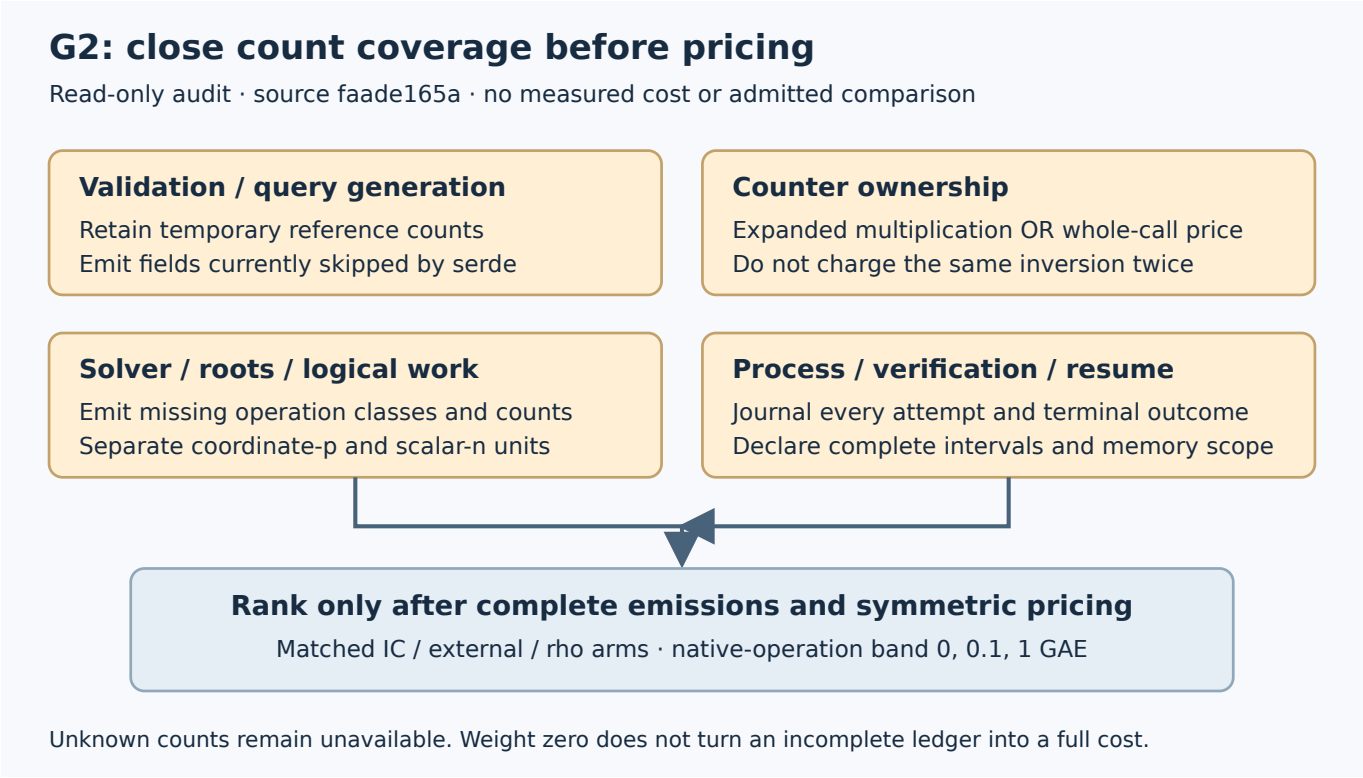
Integration and current validation

Merge a030c1630518564bdb3633815391858b424377ce joins local 45b53fbc8 and reviewed remote 0cc5e0d23. Only the append-only daily log and problem index conflicted. All 1022/1140 parent paragraphs are preserved, including 26 complete remote entry groups and the exact local prefix. All 89/91 parent problem links survive; main's hashed-ECDL update and IC campaign row and the local PKM update are retained. Local PKM source and frozen study evidence are unchanged by the merge.

Copilot cannot access unpublished local history; the cannot-act fallback and exact scope were recorded on PR2949 in comment 6530542078. No rebase or force push. All earlier inputs/receipts remain intact. The merge is reviewed, not automatically validated by its parents.

The changed Lean gate now passes 18 declarations with zero trusted exceptions. Official module kernel replay passes against imports, not –fresh. Fresh native validation stopped in its first compilation with Cargo/supervisor101: an unclosed delimiter in binary_semaev_chain_sat.rs, byte-identical at this merge, PR head0cc5 and main parentd4b3. Zero tests ran; eight later phases were not reached. Exact small logs, source comparison and terminals are retained in validation/integration-0cc5/native-failure/. No stale binary or earlier23-pass receipt is treated as validation of this source. No test or gate was waived. Current main's source preview contains a repair, but a complete build is unverified and the PR now reports conflicts; Copilot-first resolution is required.

Earlier 23-example and capped-library receipts apply to faade only. The repaired integration needs its own actual phase outcomes, binary hashes and final library reconciliation. Local commits remain unpushed pending that validation. Required gate,size,artifacts and applicable reviews must pass on the exact published head before guarded merge; auto-merge remains paused.



Accounting work still required

Read-only recheck at a030 confirms the G2 gaps: temporary affine validation counts are dropped, query GroupOps field details are skipped by serialization, and inversion calls expand through already-counted multiplication helpers. Future pricing must avoid charging both a whole call and its expanded children. F4/root/logical counts remain partial; scalar-n and coordinate-p arithmetic need distinct class/calibration identities. Durable failed-call journaling and full process/replay/resume coverage remain open.

The next source change should retain/emit those counts and declare ownership with meaningful native conservation/refusal tests. The audit is not the implementation. Missing counts cannot be treated as zero, including at the zero-weight pricing endpoint. The symmetric 0/0.1/1 GAE band and matched external/rho references remain required. No goal threshold or target law changes.

Sources, formal scope and execution limits

PKM 2016 Algorithm 1/§3.2 already accounts for relation dependence and linear algebra; symmetry and provenance certificates are established ideas. Primary source: people.maths.ox.ac.uk/petit/files/16PKC_primeECDLP.pdf. Repository AGENTS §5a and docs/ic/measurement/README.md define complete emissions and comparable units. Actual source audit paths are listed in G2_ACCOUNTING_AUDIT.md and G2_RECHECK_A030.md.

The five supporting Lean contracts construct cyclic scalar coordinates from exact group/cardinality hypotheses and prove original-row recovery/kernel implications. Rust/Mathlib representation, polynomial correspondence and complete solving/extraction-cost bridges remain unproved. Standard proof foundations only; no new complexity theorem is claimed.

No checkpoint/artifact/lab identity or qualified source-bound native runtime/resume route is configured. No scientific launch or funding, lease, IAM, quota or pool change occurred. The full-panel card stays blocked/unfunded/unimplemented at its historical pin. Existing canonical IC/curve/performance graphs are unchanged because no admitted metric or new edge is added. This explanatory diagram maps accounting obligations; no performance comparison was measured.

Evidence: validation/native-final-faade/receipt.json, validation/integration-0cc5-resolution.json and validation/integration-0cc5/proof-receipt.json. Current native compilation failed101; downstream phases are not reached, not inferred passes. Software integration or passing CI does not complete G1–G6.